

Jane explained that she would build a React component using a custom `useInterval` hook to create a timer that increments every second, with Start, Pause, and Reset buttons for control. While her implementation uses React with TypeScript, the interviewer was primarily evaluating her grasp of advanced JavaScript concepts—such as closures, higher-order functions, private variables (via `useRef`), immutability patterns, and ES2015+ features.

She wrote the `useInterval` hook, shown in listing 3.12, adapted from Dan Abramov's blog post "Making setInterval Declarative with React Hooks" (https://overreacted.io/making-setinterval-declarative-with-react-hooks). This hook simplifies interval management by ensuring the callback always references the latest state or props and by providing proper cleanup to prevent memory leaks. The code can be found at `hooks/use-interval/useInterval.ts`.

Listing 3.12 Implementation of the useInterval custom hook

```
import { useEffect, useRef } from 'react';
export function useInterval(callback: () => void, delay: number |
  const savedCallback = useRef<() => void>(null);
```